# Flexigo – High-Performance EV Mobility Ecosystem

Enterprise Technical Product Requirement Document (PRD) & Engineering Scope**Author:** Lead Developer / Development Team **Version:** 4.2 **Status:** Comprehensive Technical Submission for Client Review

---

# 1. Project Overview & Technical Vision

As the designated development partner for Flexigo, our mandate is to engineer a high-availability, scalable digital infrastructure capable of managing thousands of IoT-enabled electric vehicles. This document serves as the definitive technical blueprint, outlining the transition from hardware telemetry to a consumer-facing subscription economy.

## 1.1 Engineering Core Pillars

- **Hardware-Software Synergy:** Implementation of a high-throughput MQTT broker system (e.g., EMQX) for sub-second vehicle telemetry ingestion, state synchronization, and remote command execution with Guaranteed Message Delivery (QoS 1/2).
- **Financial Integrity:** A double-entry ledger database system for wallet transactions, ensuring ACID compliance and 100% reconciliation between geospatial ride distance, battery discharge curves (SoC consumption), and automated billing events.
- **Operational Automation:** Eliminating manual bottlenecks through AI-driven KYC (Optical Character Recognition & PAD), automated predictive maintenance alerts, and programmatic franchise settlement cron jobs.

## 1.2 High-Level System Architecture

```
graph TD
    subgraph Edge Layer
        A[EV IoT Controller] <-->|MQTT over TLS| B(MQTT Broker EMQX)
        A -.->|BLE Fallback| C[Rider Mobile App]
    end

    subgraph API Gateway & Load Balancing
        D[NGINX / AWS ALB]
    end

    subgraph Microservices Core
        E[IoT Telemetry Service]
        F[User & Identity Service]
        G[Billing & Ledger Service]
```

```
        H[Fleet Ops Service]
    end

    subgraph Data Persistence Layer
        I[(PostgreSQL - Relational)]
        J[(TimescaleDB - Time-Series)]
        K[(Redis - Caching/PubSub)]
    end

    B <--> D
    C <-->|HTTPS/REST/WSS| D
    O[Admin Dashboard] <-->|WSS/REST| D
    P[Franchise App] <-->|HTTPS| D

    D --> E & F & G & H
    E --> J
    E <--> K
    F --> I
    G --> I
    H --> I & K
```

---

# 2. Stakeholder Architecture & Interaction Models

## 2.1 Rider Mobile Ecosystem

- **Target:** Delivery professionals and urban gig-workers.
- **Technical Focus:** Minimal battery drain (GPS optimization), offline-first state management using Redux-Persist & SQLite, and biometric enclave security (FaceID/TouchID).
- **Key Modules:**
    - **Identity Vault:** Encrypted local storage of KYC status, JWT rotation tokens, and digital agreements.
    - **Live Garage:** Real-time WebSocket-driven visualization of true State of Charge (SOC) and projected range.
    - **Smart Key:** Bluetooth Low Energy (BLE 5.0+) integration utilizing a challenge-response cryptographic handshake for vehicle operation in areas with zero cellular connectivity.

## 2.2 Franchise Management Interface

- **Target:** Regional Hub Managers and Fleet Attendants.

- **Technical Focus:** High-speed edge-processed QR scanning, multi-user role management (RBAC), and real-time inventory synchronization.
- **Key Modules:**
    - **Handover Protocol:** Mandatory image-upload buffer (pre-signed S3 URLs) for vehicle condition audits to prevent main thread blocking.
    - **Inventory Logic:** Dynamic and deterministic state machine tracking (Available, Assigned, In-Transit, In-Service, Quarantined) to prevent invalid operational states.

```
stateDiagram-v2
    [*] --> Available
    Available --> Assigned : Rider Subscription Active
    Assigned --> In_Transit : IoT Handshake / Unlock Command
    In_Transit --> Return_Audit : End Session Triggered
    Return_Audit --> Available : Passed Audit
    Return_Audit --> Quarantined : Failed Audit / Damage Reported
    Available --> In_Service : Scheduled Maintenance
    Quarantined --> In_Service : Repair Approved
    In_Service --> Available : Maintenance Completed
```

## 2.3 Super Admin Console

- **Target:** Central Operations and Executive Leadership.
- **Technical Focus:** Aggregated MapReduce data visualization, remote fleet immobilization, and global financial governance.
- **Key Modules:**
    - **Command Center:** Live WebSocket-driven / SSE map clusters tracking active leases bounding boxes.
    - **Policy Engine:** Global hot-reloadable config management for dynamic pricing surges, geofencing strictness, and late-fee logic.

---

# 3. Engineering Workflows & Standard Operating Procedures (SOPs)

## 3.1 Onboarding & Compliance Pipeline

To minimize drop-offs while maintaining security, the onboarding flow is architected as follows:

- **Lead Capture:** OTP-based mobile verification with device-fingerprinting to prevent duplicate accounts.
- **Automated KYC:**

    - Integration with OCR engines to extract DL and Aadhaar data.
    - Liveness detection API to ensure the user is physically present during registration.
- **Legal Enforcement:** Programmatic generation of PDF rental agreements with embedded digital signatures (Aadhaar-based eSign).

## 3.2 Asset Lifecycle & IoT Telemetry

**Session Start:** Verification of subscription validity → Command sent to IoT module via MQTT → Hardware unlock.

The following sequence highlights the sub-second authorization required for hardware unlock:

```
sequenceDiagram
    participant App as Rider App
    participant API as Flexigo Backend
    participant MQTT as EMQX Broker
    participant TCU as EV Controller (IoT)

    App->>API: Request Unlock (VIN, Geo-Coordinates)
    API-->>API: Validate Subscription & Wallet Balance
    API-->>API: Check Geofence Constraints
    API->>MQTT: Publish {"cmd": "UNLOCK", "sig": "..."} to
flexigo/v1/vehicles/{VIN}/rx
    MQTT->>TCU: Push Message (QoS 1)
    TCU-->>TCU: Cryptographic Signature Verification
    TCU-->>TCU: Disengage Hardware Immobilizer
    TCU->>MQTT: Publish ACK to flexigo/v1/vehicles/{VIN}/tx
    MQTT->>API: Webhook / Subscription match
    API->>App: Session Started
```

**In-Ride Telemetry:**

- **Critical Pings:** GPS, SOC, and Speed sent and pushed to TimescaleDB every 10 seconds.
- **Health Pings:** Controller temperature and battery cycle data sent and evaluated every 5 minutes.

**Automated Return Logic:** Post-ride handshake checks vehicle location within the geofenced Hub and triggers the "Return Audit" workflow for the Franchise Manager.

## 3.3 Immutable Financial Ledger System

To ensure mathematical guarantees on all monetary and distance-based events, Flexigo utilizes a rigorous double-entry accounting structure mapped directly to IoT events:

```
erDiagram
    WALLET ||--o{ LEDGER_ENTRY : "contains"
    USER ||--|| WALLET : "owns"
    RIDE_SESSION ||--o{ LEDGER_ENTRY : "triggers"
    RIDE_SESSION ||--|{ TELEMETRY_POINT : "generates"

    USER {
        uuid id PK
        string kyc_status
    }
    WALLET {
        uuid id PK
        decimal current_balance
    }
    RIDE_SESSION {
        uuid id PK
        timestamp start_time
        timestamp end_time
        decimal total_distance_km
    }
    LEDGER_ENTRY {
        uuid id PK
        uuid wallet_id FK
        string transaction_type "e.g., RIDE_DEBIT, TOPUP"
        decimal amount
        string idempotency_key
    }
    TELEMETRY_POINT {
        uuid id PK
        uuid session_id FK
        float gps_lat
        float gps_lng
        int battery_soc
    }
```

---

# 4. Detailed Functional Workflows & User Journeys

## 4.1 Rider Application Flow

```
flowchart TD
    Start([START]) --> App[Rider Downloads / Opens App]
    App --> Mobile[Mobile Number Registration]
    Mobile --> OTP{OTP Verification}
    OTP -- Invalid --> RetryOTP[Retry]
    RetryOTP --> OTP
    OTP -- Valid --> Profile[Create Rider Profile<br/>Name, Address, DOB -
Optional]
    Profile --> KYC[Upload KYC Documents<br/>Aadhaar mandatory, PAN/DL
optional]
    KYC --> DigiLocker[Aadhaar Validation & DigiLocker<br/>via Govt Server]
    DigiLocker --> Selfie[Upload Selfie / Face Verification]
    Selfie --> BGV[Background Verification<br/>Police/Third-party - Optional
skip]
    BGV --> VehReq{Vehicle Allocation Required?}
    VehReq -- YES --> Assign[Assign EV Scooter]
    VehReq -- NO --> OwnVeh[Rider uses own vehicle]
    Assign --> Terms[Training & Terms Acceptance<br/>Optional E-sign]
    OwnVeh --> Terms
    Terms --> AdminReview{Admin KYC Review<br/>Auto-approve in 2 mins}
    AdminReview -- Rejected --> Notify[Notify Rider]
    Notify --> KYC
    AdminReview -- Approved --> Active[Rider Account Activated]
    Active --> Plans[Access Fleet & Plans<br/>Daily/Weekly/Monthly]
    Plans --> Payment[Select Plan & Payment<br/>UPI/Card/NetBanking]
    Payment --> PmtNotif[Payment Notification to Admin]
    PmtNotif --> Alloc[Vehicle Allocation & Hub Location<br/>via WhatsApp/SMS]
    Alloc --> ActiveRide((Active Ride Phase))

    ActiveRide -.-> CancelRide[Cancel Ride Option]
    CancelRide --> CancelNotif[Admin Notified + Dues Calculated]
    CancelNotif --> Handover

    ActiveRide --> Handover[Rider Handovers Vehicle at Hub]
    Handover --> OpsCheck[Manager Checks Vehicle<br/>Uploads Photos & Collects
Dues]
    OpsCheck --> End([Ride End])
```

**Rider UI/UX Must-Haves:**

- Rider should see nearby battery charging/swap stations on the live map.

**Specific API Integrations for the Rider App:**

- **Government APIs:** DigiLocker API (Aadhaar validation), VAHAN API (e-challan lookup, notifications to admin & rider).
- **Battery Vender APIs:** Real-time battery data (stats, temperature, remaining SOC).
- **Tracking Services:** Vehicle tracking linked by Rider's mobile number.

## 4.2 Admin Panel Flow & Connectivity

The Admin Panel serves as the central command to oversee the Rider Application, approve user states, manage finances, and coordinate with franchises/hubs.

**Connection with Rider App:**

- **KYC Linkage:** When Rider submits KYC, it lands in the Admin KYC Review queue (or is auto-approved, alerting the admin).
- **Financial Linkage:** Rider plan selections and payments send immediate push notifications and ledger updates to the Admin.
- **Ride Action Linkage:** Rider ride-cancellation triggers instant notifications to Admin to reconcile possible dues.
- **Telemetry Linkage:** Admin dashboard consumes the real-time location and VAHAN e-challan alerts tied to the active rider.

**Admin User Flow:**

```
flowchart TD
    Start([START]) --> Login[Admin Login 2FA/OTP]
    Login --> Dash[Dashboard Overview<br/>Stats, Revenue, Alerts]

    Dash --> Notifications{Notifications Queue}
    Notifications --> KYC[KYC Approvals<br/>Manual Overrides]
    Notifications --> Pmt[Payment Notifications]
    Notifications --> Cancel[Ride Cancel & Dues]

    Dash --> Fleet{Fleet & Rider Management}
    Fleet --> RiderView[View Profiles & Live Tracking]
    Fleet --> Battery[Monitor Battery Data APIs]
    Fleet --> Vahan[Review VAHAN e-challans]

    Dash --> Franchise{Franchise/Hub Oversight}
    Franchise --> AssignInv[Assign EV Inventory to Hubs]
    Franchise --> MonitorHandover[Monitor Handovers & Checklists]

    Dash --> Config{Configuration & Disbursals}
    Config --> ModifyPlans[Modify Subscription Plans]
```

```
    Config --> Settle[Manage Franchise Settlements]
```

## 4.3 Admin Panel Sidebar Elements

1. **Dashboard:** High-level metrics, active leases, revenue graph, map clusters, and alerts.
2. **Rider Management:** User list, detailed profiles, KYC status queue, wallet balances, ride history.
3. **Fleet Management:** Vehicle registry (VINs), real-time map tracking, battery/SOC status, maintenance logs.
4. **Franchise & Hub Operations:** List of hubs/franchises, inventory allocated, Operation Manager profiles.
5. **Financials & Billing:** Revenue reports, payment gateway logs, rider deposit tracking, settlement to franchises, VAHAN e-challan ledger.
6. **Subscription Plans:** CRUD operations for daily/weekly/monthly pricing and deposit tier settings.
7. **Support & Tickets:** Ride cancellations, accident reports, background verification tracking, roadside assistance.
8. **Settings & Integrations:** VAHAN API keys, Battery Vendor API keys, DigiLocker Auth configs, Admin Roles & Permissions (RBAC).

## 4.4 Franchise Role & Operations Flow

The Franchise (Hub / Operation Manager) acts as the physical touchpoint between the digital platform and the physical vehicle assets.

**Connection with Rider App & Admin:**

- **Rider Handoff:** Receives physical vehicle handover from Rider.
- **Admin Sync:** Uploads audits that directly update the vehicle's state in the Admin Panel (from "In-Transit" to "Available" or "Quarantined").

**Franchise Operation Manager Flow:**

```
flowchart TD
    Start([START]) --> Login[Manager logs into Franchise App / Portal]
    Login --> Inv[Manage Hub Inventory<br/>View arriving Riders & vehicles]
    Inv --> HandoverInitiated[Handover/Return Process Initiated]
    HandoverInitiated --> Inspect[Physically Inspect Vehicle]
    Inspect --> Photos[Upload Photos - Front, Back, Left, Right]
    Photos --> Checklist[Condition Verification Checklist<br/>Damage, Tires,
Scratches]
    Checklist --> BatteryCheck[Record Battery & Charger Status]
    BatteryCheck --> DuesCheck{Check End-of-Ride Dues}
    DuesCheck -- Dues Pending --> Collect[Collect Payment via App QR]
```

```
    Collect --> MarkRet[Mark Vehicle as Returned]
    DuesCheck -- No Dues --> MarkRet
    MarkRet --> Sync[Vehicle State Syncs to Admin<br/>Available or Requires
Maintenance]
    Sync --> End([END])
```

---

# 5. Deep-Dive Feature Matrix

## 5.1 Admin Control Features

- **Geofencing Suite:** Define polygonal boundaries on a map. Automatic immobilization or edge-computed speed-governing if a vehicle exits the zone without network reliance.
- **FOTA (Firmware-Over-The-Air):** Centralized dashboard to push delta firmware updates to IoT controllers across the fleet, with automatic rollback on boot failure.
- **Fraud Detection:** Alerts for "Phantom Movement" (vehicle moving without an active session) or "Battery Mismatch / Telemetry Blackouts" indicating hardware tampering.
- **Financial Reconciliation:** Distributed cron routines generating daily automated settlement reports for franchise commissions and platform fees.

## 5.2 Franchise Operational Features

- **Resource Management:** Track battery swap station levels, charger availability, and load-balancing instructions.
- **Maintenance Logs:** Digital service records linked to vehicle VIN, including tracking the lifecycle and consumption of individual spare parts.
- **Performance Analytics:** Local hub metrics including turnover time (time from vehicle return audit to next pickup).

## 5.3 Rider Experience Features

- **Dynamic Range Estimation:** Ephemeral range calculated by ML models based on rider weight, terrain data, and historical battery discharge performance.
- **Subscription Management:** Micro-transaction capabilities to securely upgrade/downgrade plans or "Pause" subscriptions for a small fee.
- **Community Support:** Integrated ticketing system with S3 multi-part image/video upload for roadside assistance.

---

# 6. Proposed Technology Stack & Infrastructure

## 6.1 Backend & Middleware

- **Framework:** Node.js with NestJS (TypeScript) for strict type safety, dependency injection, and modularity.
- **Communication:** MQTT (via Mosquitto or EMQX clustered deployment) for IoT; WebSockets/gRPC for real-time dashboard updates and inter-service calls.
- **Search/Analytics:** Elasticsearch clustered indexing for high-speed filtering of millions of ride logs.

## 6.2 Frontend & Mobile

- **Rider App:** React Native (Expo bare workflow) for near-native 60fps performance on both iOS and Android.
- **Web Portals:** React 18 with Vite for optimized build times and Tailwind CSS combined with shadcn/ui for scalable component styling.

## 6.3 Database & Storage

- **Relational:** PostgreSQL (Amazon RDS Multi-AZ) for user, transaction, wallet ledger, and subscription data.
- **Time-Series:** InfluxDB or TimescaleDB for cleanly storing high-volume GPS and battery telemetry datasets.
- **Media:** AWS S3 (with CloudFront CDN routing) for secure storage of KYC documents and vehicle inspection photos.

---

# 7. Security & Data Integrity Protocols

| Risk Category | Technical Implementation Strategy |
|---|---|
| **Identity Theft** | JWT-based authentication (RS256) with refresh token rotation and biometric re-auth mandated for high-value wallet actions. |
| **Asset Recovery** | Independent secondary GPS/GSM tracker backup on a discrete power supply; hardware watchdog timer triggering remote ignition kill-switch logic. |
| **Data Privacy** | PII (Personally Identifiable Information) encryption at rest using AES-256-GCM; automated GDPR/DPDP compliant data purging cron-jobs. |
| **Payment Integrity** | Distributed ACID locks (e.g., Redlock) and Webhook-based idempotency keys to handle asynchronous payment gateway responses. |

---

# 8. Implementation Roadmap

The following Gantt chart models the sequential and parallel execution of our overarching engineering milestones.

```
gantt
    title Flexigo Engineering Delivery Schedule
    dateFormat  YYYY-MM-DD
    axisFormat  %M

    section Phase 1: Core
    Cloud Infrastructure (IaC)     :a1, 2024-05-01, 14d
    Identity & Security Service    :a2, after a1, 14d

    section Phase 2: Operations
    IoT Broker & Telemetry Engine  :b1, 2024-05-29, 21d
    Franchise Audit Workflows      :b2, after b1, 7d

    section Phase 3: Fintech
    Wallet & Immutable Ledger      :c1, 2024-06-26, 14d
    Payment Gateway Integration    :c2, after c1, 14d

    section Phase 4: Scale
    AI Fraud & Geofencing Rules    :d1, 2024-07-24, 14d
    Load Testing & Optimization    :d2, after d1, 14d
```

- **Phase 1: Foundation & Core Logic**
    - Infrastructure setup (VPC, RDS, MQTT Broker).
    - User Authentication & Automated KYC Integration.
    - Development of basic Admin/User CRUD operations.
- **Phase 2: Mobility & Fleet Operations**
    - IoT Protocol Implementation (Bidirectional Handshake logic between bike and server).
    - Subscription Billing Engine, Immutable Ledger, & Payment Gateway integration.
    - Franchise "Handover & Return" audit workflows.
- **Phase 3: Scaling & Optimization**
    - Advanced OpenTelemetry Analytics & Forecasting dashboards.
    - Geofencing rules and Automated Theft Prevention protocols.
    - Third-party API integrations (Delivery partner data sharing).

---

# 9. Development KPIs & Quality Assurance

- **API Latency:** 95th percentile (P95) of synchronous HTTP requests to resolve under 200ms.
- **IoT Reliability:** 99.9% success rate for end-to-end remote unlock commands resolving within 2-3 seconds.
- **Billing Accuracy:** Zero-tolerance fault policy for discrepancies between recorded geospatial ride time/distance and the billed amount in the distributed ledger.

---

# 10. Conclusion

This document serves as the final technical agreement for the Flexigo platform. Our engineering team is focused on delivering a system that is not only functional but resilient to the unique challenges of the EV mobility space. By centralizing operations through this tech stack, we ensure that the business remains agile and data-driven at every stage of growth.

**Next Steps:**

- Technical sign-off of the PRD by the client / Architecture Review Board.
- Finalization of the UI/UX Design System workflows.
- Initialization of the CI/CD pipeline and Sprint 1 development.